

# Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

**12 V**

6 cells

**12 Ah**

20 h to 10.5 V at 25°C

**AGM**

Valve-regulated lead-acid

## Electrical performance

|                       |                                                |
|-----------------------|------------------------------------------------|
| Rated capacity        | 12 Ah at 20 h to 10.5 V, 25°C                  |
| 10-hour capacity      | 11.2 Ah to 10.5 V at 25°C                      |
| 1-hour capacity       | 7.2 Ah to 9.60 V at 25°C                       |
| Internal resistance   | 19 mohm at 25°C                                |
| Maximum discharge     | 180A for 5 seconds at 25°C                     |
| Self-discharge        | <=3% per month at 25°C                         |
| Design life / standby | 5 years (JIS, 25°C); 3-5 years (Eurobat, 20°C) |

## Charging and operating conditions

|                               |                                                            |
|-------------------------------|------------------------------------------------------------|
| Charge method                 | Constant voltage                                           |
| Float / cycle voltage         | 13.6V-13.8V / 14.5V-15.0V at 25°C                          |
| Maximum charge current        | 3.6A                                                       |
| Temperature compensation      | Float: -18 mV/°C per battery; cycle: -30 mV/°C per battery |
| Operating temperatures        | Discharge: -15 to +50°C; charge: -10 to +50°C              |
| Storage temperature           | -20 to +50°C                                               |
| Nominal operating temperature | 25 ±3°C                                                    |

## Mechanical specification

|                        |                                                   |
|------------------------|---------------------------------------------------|
| Dimensions (L × W × H) | 151±1.5 × 98±1 × 95±1 mm; overall height 101±1 mm |
| Approximate weight     | 3.40 kg ±4%                                       |
| Terminal / torque      | T1/T2; N/A                                        |
| Container              | ABS, UL94-HB; V-0 optional                        |

# Construction & applications

AGM construction with high-performance plates and electrolyte for standby and general-purpose power. The Yellow HR/HRL range uses reinforced grids and a special electrolyte formulation for UPS and intensive high-current discharge.

## Battery construction

|                 |                   |
|-----------------|-------------------|
| Positive plate  | Lead dioxide      |
| Negative plate  | Lead              |
| Container / lid | ABS / ABS         |
| Safety valve    | Rubber            |
| Terminals       | Copper            |
| Separator       | Glass fibre (AGM) |
| Electrolyte     | Sulfuric acid     |

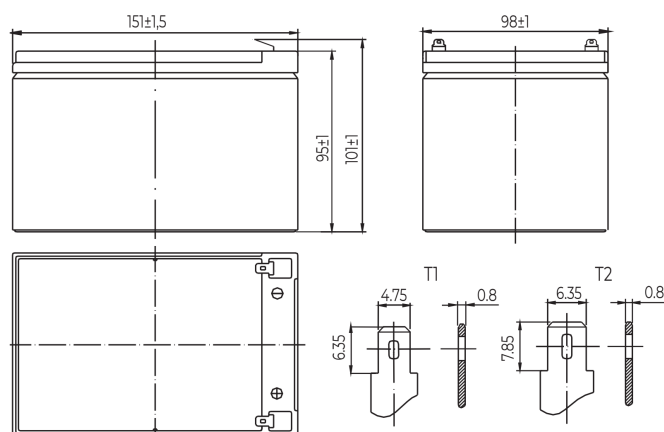
## Features

- Insulated intercell connections.
- Gas recombination efficiency up to 99%.
- High-rate discharge performance.
- Special welding process.
- Self-regulating valves; no water topping-up.
- Shelf life without recharge: 6 months at 25°C.

## Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

## Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

# Discharge performance

Fully charged battery / 25°C

## Constant-current discharge / A

| V/cell | 5m    | 10m   | 15m   | 30m   | 1h   | 2h   | 3h   | 4h   | 5h   | 10h  | 20h  |
|--------|-------|-------|-------|-------|------|------|------|------|------|------|------|
| 1.60   | 45.80 | 29.00 | 22.70 | 12.80 | 7.86 | 4.30 | 2.97 | 2.46 | 2.09 | 1.14 | 0.61 |
| 1.65   | 44.50 | 28.20 | 22.10 | 12.50 | 7.74 | 4.27 | 2.96 | 2.45 | 2.08 | 1.13 | 0.61 |
| 1.70   | 42.60 | 27.00 | 21.30 | 12.10 | 7.55 | 4.24 | 2.94 | 2.43 | 2.07 | 1.13 | 0.60 |
| 1.75   | 40.80 | 25.80 | 20.60 | 11.80 | 7.40 | 4.17 | 2.92 | 2.41 | 2.05 | 1.12 | 0.60 |
| 1.80   | 38.50 | 24.40 | 19.50 | 11.40 | 7.17 | 4.07 | 2.83 | 2.34 | 1.99 | 1.10 | 0.59 |

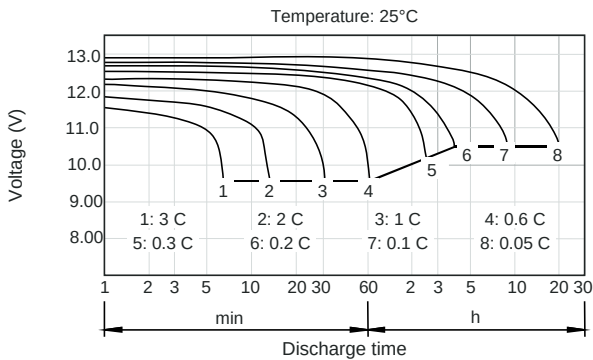
## Constant-power discharge / W per cell

| V/cell | 5m    | 10m   | 15m   | 30m   | 1h    | 2h   | 3h   | 4h   | 5h   | 10h  | 20h  |
|--------|-------|-------|-------|-------|-------|------|------|------|------|------|------|
| 1.60   | 85.33 | 54.67 | 43.17 | 24.33 | 15.17 | 8.38 | 5.88 | 4.88 | 4.17 | 2.27 | 1.22 |
| 1.65   | 82.67 | 53.00 | 42.00 | 24.00 | 14.95 | 8.33 | 5.85 | 4.85 | 4.13 | 2.27 | 1.21 |
| 1.70   | 79.33 | 50.83 | 40.50 | 23.17 | 14.57 | 8.27 | 5.82 | 4.82 | 4.12 | 2.25 | 1.21 |
| 1.75   | 75.83 | 48.67 | 39.17 | 22.67 | 14.27 | 8.13 | 5.77 | 4.78 | 4.08 | 2.23 | 1.20 |
| 1.80   | 71.67 | 45.83 | 37.00 | 21.83 | 13.83 | 7.93 | 5.60 | 4.65 | 3.97 | 2.20 | 1.18 |

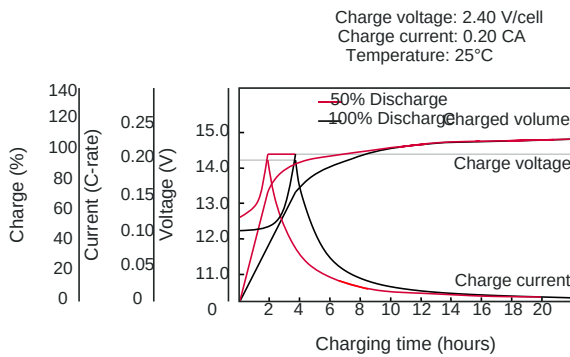
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

### Characteristic curves

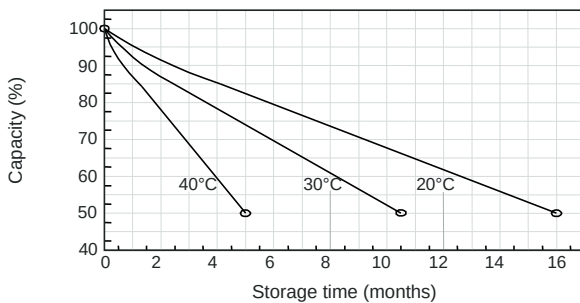
#### Discharge characteristics



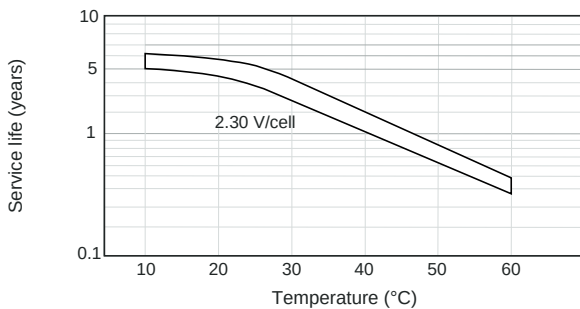
#### Charging characteristics



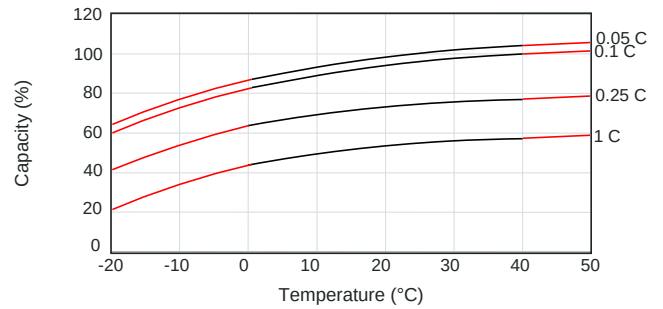
#### Self-discharge



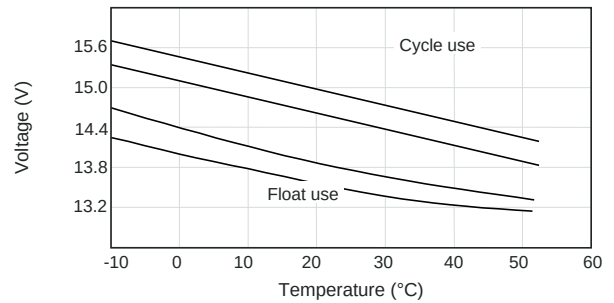
#### Float life vs. temperature



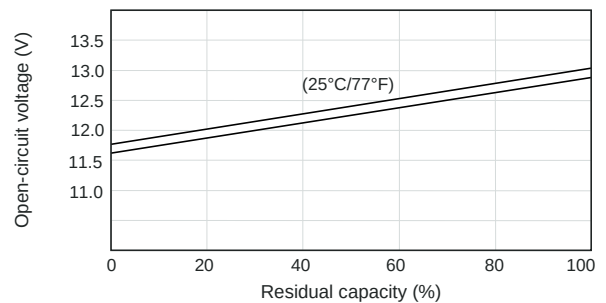
#### Capacity vs. temperature



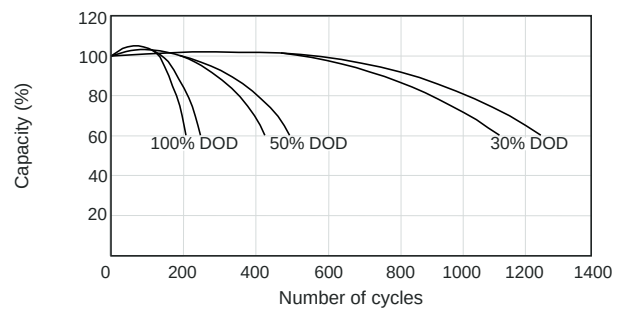
#### Charge voltage vs. temperature



#### Open-circuit voltage vs. capacity



#### Cycle life vs. depth of discharge



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.